

THOMAS MELBOURNE

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PROFESSIONAL SUMMARY

I'm a Staff Engineer with 12 years building resilient systems at scale. I've spent most of my career designing distributed architectures and mentoring engineers through complex technical problems. I enjoy getting hands-on with difficult infrastructure challenges while helping teams ship better code. Currently based in London, focused on backend systems and cloud architecture.

TECHNICAL SKILLS

Languages: Go, Python, Java, Rust, TypeScript

Cloud Platforms: AWS, GCP, Kubernetes

Databases: PostgreSQL, Redis, Elasticsearch, DynamoDB

Architecture: Microservices, Event-driven systems, Distributed systems

DevOps: Docker, Terraform, CI/CD pipelines, Prometheus, Grafana

Tools: Git, gRPC, Apache Kafka, Linux

WORK EXPERIENCE

Staff Engineer

Zenith Systems Ltd

<https://zenithsystems.io>

London, United Kingdom

June 2021 - Present

Led the redesign of our core payment processing system, reducing latency by 65 percent and improving throughput to handle 50k transactions per second with zero data loss during incidents.

Established engineering standards and conducted architecture reviews across eight teams, catching significant design issues before implementation and sharing knowledge across the organisation.

Mentored five junior and mid-level engineers through career progression, with three promoted to senior roles within the past two years based on their growth and impact.

Owned the migration of our legacy monolith to Kubernetes, running on-call rotations and incident response, reducing deployment time from six hours to twelve minutes.

Senior Engineer

ClearMetrics Ltd

<https://clearmetrics.net>

London, United Kingdom

March 2018 - May 2021

Built a real-time analytics pipeline processing two billion events daily using Kafka and Go, achieving single-digit latency for customer dashboards serving Fortune 500 clients.

Designed and implemented a distributed tracing system adopted across the entire platform, enabling teams to debug production issues 70 percent faster and reducing mean time to resolution from four hours to forty minutes.

Led incident response for critical production outages, documenting post-incident reviews and driving systemic improvements that prevented recurrence of similar failures.

Worked with product teams to define API contracts and performance requirements, delivering a gRPC-based service layer that became the foundation for three major product features.

Software Engineer

Vertex Data Solutions

<https://vertexdata.co>

London, United Kingdom

July 2012 - February 2018

Developed backend services for a data warehousing platform using Java and PostgreSQL, supporting petabyte-scale analytics for enterprise customers across financial services and retail sectors.

Optimised database query performance for slow reporting jobs, implementing strategic indexes and query rewrites that reduced execution time by 80 percent and saved customers significant infrastructure costs.

Participated in on-call rotation and responded to production incidents, building monitoring dashboards and alerting rules that increased system observability and reduced mean time to detection.

Collaborated with three-person platform team to evaluate and adopt Docker early, building container infrastructure and deployment tooling that became the foundation for company-wide containerisation.

EDUCATION

Bachelor of Science in Computer Science

University of Manchester

Manchester, United Kingdom

Graduated June 2012